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An IoT-Based System for Monitoring Electricity Usage in Student Dorm Rooms Using the ESP32

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ABSTRACT

Real-time monitoring and control of electricity usage are essential for managing power consumption, particularly in environments such as dorm rooms or rental housing. This study designs and develops an Internet of Things (IoT)-based system using an ESP32 microcontroller combined with a PZEM-004T sensor, a relay module, and an OLED display. The ESP32 was chosen because it supports Wi-Fi connectivity and offers fast data processing capabilities. The PZEM-004T sensor detects electrical parameters such as voltage, current, power, and energy (kWh). The relay module functions as an automatic switch to disconnect or connect the current to electrical devices, either automatically or via manual control through the Blynk app. Meanwhile, the OLED display shows real-time electricity consumption data for each room. This system is designed to independently monitor and display electricity usage in each room, as well as calculate estimated costs based on a predetermined rate per kWh. With the help of the Blynk app, users can view real-time data remotely and control electrical devices as needed. Each room is equipped with one sensor, one relay, and one OLED display, all connected to a single ESP32 unit, enabling efficient monitoring of multiple points simultaneously. Based on test results, the system demonstrates good performance with high accuracy and fast response times. This system is expected to help dormitory residents control their electricity usage, raise awareness of energy efficiency, and reduce wasteful electricity costs.

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1. INTRODUCTION

Electricity usage in residential settings, particularly in shared housing or boarding houses, is often not adequately monitored. This situation has the potential to lead to energy waste and result in unwanted high electricity bills for both tenants and landlords. In this context, there is a need for a real-time electricity consumption monitoring and control system that enables users to become more aware of their energy usage. Through the use of *Internet of Things* (IoT) technology, dormitory residents can monitor electricity usage directly, take energy-saving measures, and support the creation of a more efficient and environmentally friendly lifestyle. The urgency of this research lies in its contribution to improving energy efficiency, reducing electricity costs, and encouraging the adoption of smart technology in daily life.

In line with advancements in information and communication technology, the concept of the Internet of Things (IoT) presents significant opportunities to connect various physical devices so they can communicate with one another via the internet. One particularly interesting application of IoT is in the field of electricity management, especially in the residential sector and rental housing such as dormitories and shared rooms. Electricity consumption in this sector continues to rise, driving the development of

innovations to optimize energy usage [1]. IoT-based monitoring systems offer the ability not only to measure electrical parameters such as voltage, current, power, and energy, but also to provide users with more informative data, including cost estimates based on PLN rates. Thus, this system can serve as an effective solution to address energy efficiency challenges at the household and small-scale residential levels.

The issue of uncontrolled electricity consumption in boarding houses often results in additional costs for both residents and managers. The conventional electricity meter system provided by PLN only provides total usage data, without being able to provide detailed information for each room or daily usage. This situation makes it difficult for tenants to manage their electricity consumption and prevents boarding house owners from efficiently monitoring usage per tenant. Therefore, a technology-based monitoring system is needed that can provide real-time, detailed, and accurate information regarding energy consumption. In line with this need, previous studies have highlighted the urgency of implementing an Internet of Things (IoT)-based monitoring system. For example, a study by Syamsuri, Amalia, Mudjiono, and Imron designed an electricity consumption monitoring system using an ESP32 microcontroller, a PZEM-004T sensor, a , and the Blynk app for remote control. The results of this study demonstrated that the developed system could provide accurate information on electricity consumption per room while simplifying users' ability to control electrical devices [2]. A similar study by Ibrahim, Ridyandhika Riza, & Bekti Yulianti, using the NodeMCU ESP8266 with a PZEM-004T sensor and the Blynk app, showed a relatively small measurement error rate that fell below the sensor module's tolerance, making it reliable for practical applications in student housing [3].

Previous research studies have also shown that electricity management is a critical issue across various sectors, including households and institutions. For example, a study by Zainal Abidin and Aynin Churlillah conducted an energy audit of a campus building to evaluate energy consumption patterns. The results showed that although efficiency levels were quite good, the use of energy-efficient equipment was necessary to further reduce waste. The study underscored the importance of energy management strategies, both on a large scale and in smaller settings such as boarding houses, with the aim of optimizing energy use and reducing electricity costs [4]. Thus, the urgency of implementing an electricity monitoring system is relevant not only in the industrial or educational sectors but also in simpler residential environments.

As technology advances, IoT-based energy monitoring systems continue to be developed using various microcontroller components and sensors. Research by Lasera designed an electrical power monitoring device based on the Arduino Uno R3 and an Ethernet shield module that can be controlled via a smartphone using the NodeMCU ESP8266 and the Blynk app [5]. Furthermore, Adiwiranto developed a prototype system for monitoring electricity consumption and estimating costs for household appliances using a PZEM-004T sensor with an ESP8266 processor [6]. In a more specific context, Hamami built an IoT-based electricity usage cost monitoring system for boarding houses, utilizing a PZEM-004T sensor and an ESP8266 microcontroller [7]. These studies confirm that IoT integration enables detailed measurement of current, voltage, and power, and is capable of providing more transparent energy usage information to users.

Recent studies indicate a trend toward strengthening the implementation of IoT in households and boarding houses. In Firza Istigfar's study, the researchers developed an Arduino-based system for controlling and monitoring electricity consumption using the Blynk app and the PZEM-004T sensor , with results displayed on an LCD screen [8]. Meanwhile, Amirah designed an electricity monitoring system for residential customers using a combination of an Arduino Uno R3, an ESP8266, an ACS712 current sensor, and a ZMPT101B voltage sensor. The results of these studies confirm that the use of IoT-based microcontrollers and sensors can provide more informative, accurate, and easily accessible real-time electricity consumption data for users [9]. This indicates that the direction of energy monitoring system development is increasingly focused on improving accuracy, ease of access via mobile devices, and the affordability of implementation in everyday environments, including boarding houses.

Building on previous research, this study focuses on the design and implementation of an IoT-based electrical energy consumption monitoring system using the ESP32 microcontroller. The ESP32 was chosen for its built-in Wi-Fi connectivity, efficient data processing capabilities, and broad support for various sensors and modules. The PZEM-004T is used to measure key electrical parameters, while the Blynk app serves as the user interface for monitoring and controlling the device via a smartphone. This study is limited to the implementation of an electricity monitoring system in a dorm room, covering

hardware and software design, system testing, as well as analysis of electricity consumption data and cost estimation.

The objective of this study is to identify patterns of electricity consumption in student housing units and to understand how an IoT-based monitoring system using the ESP32 operates. The findings of this study are expected to provide practical benefits, particularly for landlords in managing electricity bills and for residents in becoming more aware of their energy consumption. Additionally, this study can serve as an academic reference for the development of smart building and smart home technologies in simple residential settings. Thus, the implementation of an IoT-based electricity monitoring system not only contributes to energy efficiency but also aligns with the trends of digitalization and automation in modern life.

2. METHOD (10 PT)

This study is a quantitative experimental study, conducted through several interconnected stages [10]. The initial stage is a literature review involving the collection of information from scientific journals and reference books, specifically regarding the ESP32 microcontroller, the PZEM-004T sensor, the relay module, and the OLED display. The results of this literature review served as the foundation for designing the electrical energy monitoring system. The next stage involved hardware design tailored to the components' functions, followed by software development using the Arduino IDE to program the ESP32 to control and integrate all components.

The next step is system integration, which involves combining the hardware and software that have been developed into a single unit. Once the system is integrated, testing and analysis are conducted to ensure the system functions as intended. Testing focuses on sensor reading accuracy, response speed, and control capabilities via the application. The test results are then analyzed to evaluate the system's performance and its alignment with the research objectives. Thisresearch workflow is visualized in a flowchart for easier understanding

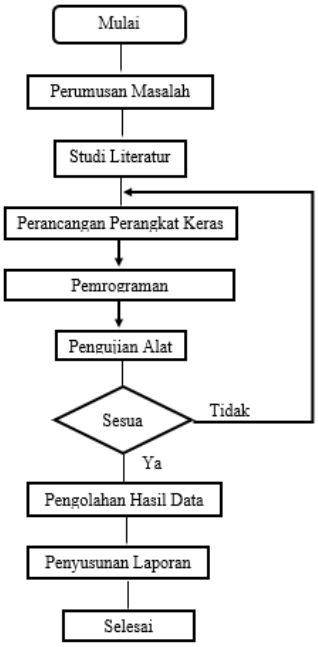


Figure 1: Research Flowchart

Before proceeding to the hardware and software implementation, a system block diagram was first designed to illustrate the relationships between components and the overall system workflow. This block diagram explains the main functions of each part, starting from the PZEM-004T sensor that reads electrical parameters, the ESP32 as the control center and data processor, the OLED display as the information display,

to the relay that controls the flow of electricity. With this block diagram, system design becomes more focused and easier during the implementation phase.

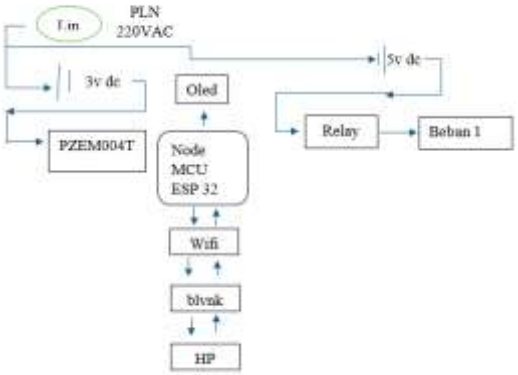


Figure 2: Circuit Block Diagram

The sensor system design utilizes the PZEM-004T module, which measures voltage, current, power, power factor, and electrical energy in real time. This module is equipped with integrated voltage and current (CT) sensors, simplifying implementation. The measurement data is sent to the ESP32 microcontroller for further processing. The ESP32 is programmed using the Arduino IDE, utilizing the built-in WiFi capabilities ( ) to send data to the Blynk app. Thus, the ESP32 acts as a control center that processes information from the sensors and presents it directly to the user via a mobile device.

The output system is designed to display measurement data and control electrical current. An OLED display is used to show information such as voltage, current, power, and energy, which is displayed in real-time after being processed by the ESP32. Additionally, a relay module serves as an electronic switch controlled by the ESP32, either automatically or manually via the Blynk app. By combining the OLED as a data display and the relay as an electrical current controller, the system is capable of providing both monitoring and control functions for electrical energy usage in a practical and efficient manner.

3. RESULTS AND DISCUSSION (10 PT)

The results of this study will summarize the testing and discussion of an IoT-based electricity consumption monitoring system control device using an ESP32 microcontroller. Testing was conducted on each major component, including the switching adapter, LM2596 step-down module, ESP32, Blynk app, PZEM-004T sensor, OLED display, and relay. Furthermore, comprehensive testing of the entire system was performed to ensure proper system integration. The purpose of these tests is to evaluate the performance of each component while ensuring the system as a whole functions as designed—that is, capable of accurately and efficiently monitoring and controlling electricity consumption in real-time.

The switching adapte*r* was tested to determine the output voltage to be used as the power source for the microcontroller. The equipment used included a 12V 3A switching adapter, a multimeter, and a data sheet. The testing procedure involved connecting the adapter to a 220VAC power source, then placing the black probe on GND and the red probe on the adapter's output to read the results on the multimeter.

Table 1: *Switching Adapter* Measurement Results

Input (V) AC	Output (V) DC	Description
220 V	12.0 V	Correct

Source: Research practice, 2025

Test results show that the switching adapter produces an output voltage of 12.0 VDC. This value meets the required specifications, so the adapter can be considered to be functioning normally.

Testing of the LM2596 *step-down* module was conducted to ensure that the module functions properly as a voltage regulator, converting 12VDC from an adapter to 5VDC. The equipment used included the LM2596

step-down module, *an adapter*, and *a multimeter*. The procedure was performed by connecting the LM2596 input to the adapter according to the positive and negative terminals ( ), then placing the black multimeter probe on the negative terminal and the red probe on the module's positive output. The output voltage value was then observed on the multimeter.

Table 2: LM2596 *Step-Down* Results

Input (V) AC	Output (V) DC	Description
12.0 V	5.0 V	Correct

Source: Research practice, 2025

Test results show that the LM2596 *step-down* module produces an output voltage of 5.0 VDC from *a* 12.0 VDC *input*. This value meets the system's requirements, so the module can be deemed to be functioning normally.

ESP32 testing was conducted to ensure the microcontroller functions normally and can be programmed using the Arduino IDE. The equipment used included a 5V power supply, a USB Type-C data cable, and an ESP32 microcontroller.



Figure 3: ESP32 Testing

The test results show that the ESP32 is in normal condition, indicated by the indicator LED being lit and the program being able to be uploaded and run as needed. This proves that the microcontroller is ready to be used as the control center in the monitoring system.

The Blynk app was tested to ensure that command transmission and reception functions were working normally, as well as to test the *Overload Alert* feature, which sends an automatic notification if power exceeds 250 watts. The test used a smartphone, an ESP32, a USB Type-C cable, and a laptop. The procedure involved connecting the ESP32 to the laptop, uploading the program, connecting to Wi-Fi, and then running the Blynk app to check for notifications and data transfer.

Table 3: Test Results for Power Overload Notifications

Room	Power (W)	Notification Status
1	180	Not sent
2	255	Sent
3	220	Not sent
4	310	Sent

Source: Research practice, 2025

Table 4: Results of Data Transfer Duration to Blynk

Command	Internet connection

	Three (seconds)	Smartfren (seconds)	Wi-Fi (seconds)
Sending a command to turn on	3.15	3.40	2.23
Sending a command to turn off	4.12	4.25	3.18

Source: Research practice, 2025

Test results show that the system is capable of sending notifications when power consumption exceeds the specified threshold. Additionally, command response speed is significantly influenced by the quality of the internet connection used. The " " screen from the Blynk app test can be seen in the following image.

Testing of the PZEM-004T sensor was conducted to ensure that the module functions properly when reading various electrical parameters, such as voltage, current, active power, energy, frequency, and power factor. The equipment used in this test included the PZEM-004T sensor, an ESP32, a 0.96-inch I2C OLED display, a 5V power supply, a laptop, a USB Type-C cable, jumper wires, and a multimeter as a reference tool. The test began by uploading the program to the ESP32 via the Arduino IDE, then connecting the ESP32 to the PZEM-004T sensor module. Once the system was connected, an electrical load was applied to observe the values displayed on the OLED and the Blynk app.

The testing process continued by comparing the sensor readings with the values obtained from the multimeter. This comparison aimed to determine the accuracy of the PZEM-004T sensor when used under real-world conditions. Through this testing, it is hoped that the sensor module can be confirmed to provide accurate and stable data, making it reliable for use in an IoT-based electricity consumption monitoring system.

Table 5: PZEM-004T Sensor Testing

Pzem004t	Sensor Reading		
	With Load	No load	Description
Voltage (V)	232	232	Sesusi
Current (A)	0.18	0.00	According to
Power (W)	41	0	As per
Energy (kWh)	0.38	0.38	As per
Frequency (Hz)	50.00	50.00	Compliant
Power factor	0.57	0.0	Compliant

Source: Research Practice, 2025

Test results show that the PZEM-004T sensor can read all parameters accurately, both under loaded and unloaded conditions. The sensor is deemed to be functioning normally because the displayed values match the reference readings.

OLED testing was conducted to ensure the module can display data according to the created program. The OLED is used as a serial display medium to show the sensor reading values from the PZEM-004T. The equipment used includes an I2C OLED, an ESP32, a 5V power supply, a laptop, and jumper cables. The procedure was carried out by uploading the program to the ESP32 via the Arduino IDE, then connecting the

ESP32 to the OLED module. Once connected, it was verified whether the OLED displayed the data according to the program.

Table 6: OLED Testing

Condition	Data Display	Description
Off	Not displayed	On
On	Appears	As expected

Source: Research practice, 2025

Test results show that the OLED functions normally. The module displays sensor data when it is on as programmed ( ), while no data appears when it is off.

Relay testing was conducted to ensure the module functions properly when supplied with 5VDC, both in NO (*normally open*) and NC (*normally closed*) states. The equipment used included a 4-channel 5V relay, an ESP32, a power supply, and a multimeter. The procedure involved connecting the ESP32 pins to the relay pins, applying 5V to the circuit, and recording the test results.

Table 7: Relay Testing

Input Voltage v (dc)	Relay Condition	Description
0 V	No	Not lit
5V	NC	On

Source: Research practice, 2025

Test results show that the relay is in the *normally open* state when no voltage is applied. Conversely, when a 5V voltage is applied, the relay switches to *the normally closed state* and functions as designed.

A comprehensive test of the device was conducted to ensure the system operates as intended and that the circuit functions properly. The equipment used includes a 220VAC power outlet, a USB cable, a laptop with the Arduino IDE software, and a smartphone with the Blynk app. The testing procedure began by turning on the laptop, uploading the program to the ESP32 via the Arduino IDE, and then assembling the modules according to the overall schematic. After connecting to a 220VAC power source and establishing a successful Wi-Fi connection, the system was tested via the Blynk app on the smartphone.

Blynk app on the smartphone.

Table 8: Overall Testing

No	Test Section	Expected Results	Test Results
1	Testing of the LM2596 *step-down* adapter and module.	The LM2596 adapter and module can step down the voltage from 220VAC to 5VDC via switching.	The adapter circuit functions properly.
2	ESP32 Testing.	The ESP32 can store programs uploaded via the Arduino IDE and can connect to the internet.	The ESP32 circuit is functioning properly.
3	Testing the PZEM004T sensor.	The PZEM004T sensor can measure voltage, current, power, energy, frequency, and power factor.	The Pzem004t sensor circuit is functioning properly.
4	Testing the *relay* channel module.	The relay can turn on and off when receiving commands from the Blynk app.	The relay module circuit is functioning properly and operates as commanded.

No	Test Section	Expected Results	Test Results
5	Testing the 0.96-inch OLED module and I2C.	The OLED can display text and values obtained from the sensor.	The I2C LCD circuit is functioning properly.

Source: Research Practice, 2025

Test results show that all circuits function properly as expected. The adapter is capable of stepping down the voltage as needed, the ESP32 can store programs and connect to the internet, the PZEM-004T sensor works accurately, the relay responds correctly to Blynk commands, and the OLED displays sensor data normally. Thus, the entire device is ready for use as an IoT-based electrical energy consumption monitoring system.

Based on the overall test results, it can be confirmed that the system functions properly as designed. When the switch is turned on, the OLED displays an initial screen showing the text “Monitoring” along with the user’s name and student ID, followed by data from the sensors. After connecting to the network, the OLED displays the status “Connection Successful,” indicating that the system is connected to the Blynk app. The Blynk app then displays real-time sensor measurementdata and serves as a control interface for the relay to turn the power on or off as needed by the user.

The PZEM-004T sensor has been proven to accurately measure voltage, current, power, energy, frequency, and power factor. Testing was conducted by applying loads such as a fan, TV, rice cooker, laptop, and smartphone over a two-week period, and comparing the sensor’s readings with those of a multimeter. The results showed very small differences, with an average voltage error of 0.04–0.06% and an average current error of 1.96–5.88%. The measured values are displayed alternately on the OLED according to the program, while the Blynk app displays the same data and provides relay control options. Thus, this device is capable not only of monitoring but also of effectively and efficiently controlling electricity consumption.

4. CONCLUSION

Based on the test results and discussion, it can be concluded that the design of this electrical energy consumption control and monitoring device uses the ESP32 microcontroller as its main component, supported by a PZEM-004T sensor, an I2C OLED display, and a relay module. This system can be controlled and monitored remotely via the Blynk app connected to the internet. The device begins operation when the ESP32 is powered on to search for a Wi-Fi or hotspot SSID and password; once connected, the PZEM-004T sensor measures electrical parameters such asvoltage, current, power, energy, frequency, and power factor. Measurement results are displayed on the OLED screen, while the Blynk app serves as a monitoring interface and controls the relay, which turns electrical equipment on or off.

Test results show that the PZEM-004T sensor has a high level of accuracy , with an average measurement error of 1.2% for voltage and 2.1% for current, which remains within the standard tolerance limits for measuring instruments. The relay system was found to operate optimally with a response time of less than 1 second when controlled via the Blynk app. The overload alert feature also works well, capable of detecting power consumption above 250 Watts and sending automatic notifications to the app with a 100% success rate and an average response time of 1–2 seconds. Additionally, the system is equipped with an electricity cost estimation feature based on the rate per kWh, which helps boarding house residents monitor their energy usage while encouraging efficiency and conservation efforts.

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